FOODBRIDGE

Surplus Food Rescue Network

Good Food. No Waste. Greater Impact.

We connect surplus food from large gatherings with nearby NGOs and volunteers safely before collection windows close.

Team Name: Techie Titans

THE PROBLEM

Surplus exists. Connection doesn't.

Excess from preparation
Attendance varies, leaving large amounts of unused food.

Disconnected ecosystem
Donors and NGOs operate in silos, unaware of nearby surplus.

Narrow window, high risk
Delays in coordination lead to spoilage or missed pickups.

What this leads to

Food is discarded
Good opportunities are missed
Manual coordination causes delays
No records of what could have been rescued

How do we safely connect surplus food with the right recipients before time runs out?

OUR SOLUTION

One Platform. Smarter Food Rescue.

We don't just find the nearest NGO.
We deliver safety-eligible, feasible matches before deadlines.

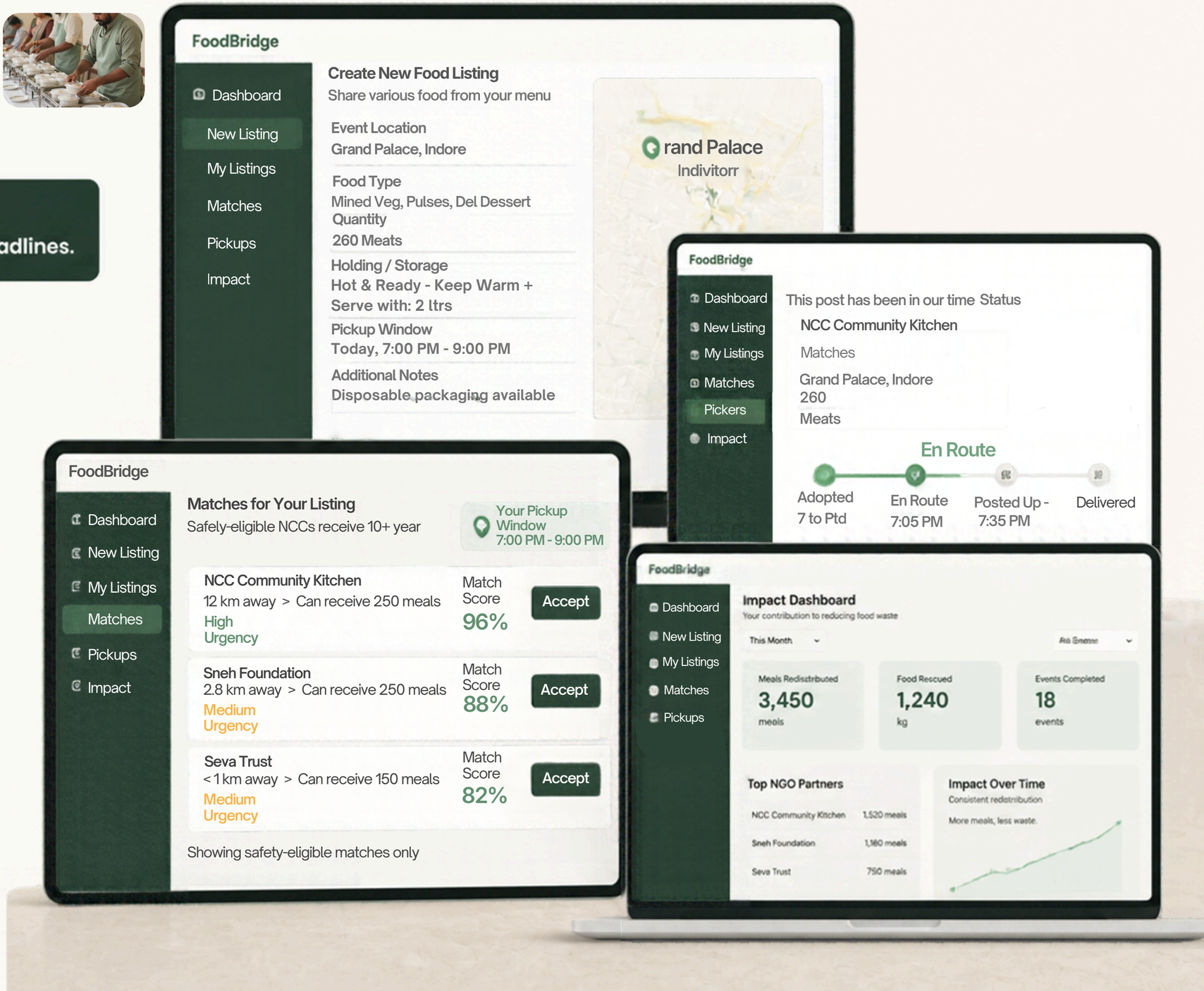
Event-Based Listings
List location, food type, quantity, handling needs, and pickup window.

Safety-First Eligibility
Our system checks safety rules and filters only eligible food.

Smart Matching Algorithm
Ranks NGOs by urgency, distance, capacity, and availability.

Alerts & Tracking
Real-time notifications, accept or decline, and live pickup status updates.

Impact Dashboard
Track meals redistributed and quantify food rescued.



OUR USP

Beyond basic food donation

FEATURE	TRADITIONAL APPROACH	FOODBRIDGE
Food discovery	Manual calls/messages	**Centralized digital listings**
Donor coverage	Personal contacts	**Weddings, bhandaras, langars, concerts**
Matching	Informal coordination	**Safety-first urgency ranking**
Time management	Manual follow-ups	**Deadline-aware alerts**
Pickup tracking	Inconsistent records	**Accepted, collected, delivered statues**
Impact measurement	Incomplete records	**Recorded meals and quantities**

SAFETY-FIRST MATCHING ENGINE

SAFETY FILTER → **URGENCY + DISTANCE + CAPACITY** → **FEASIBLE MATCH** → **PICKUP**

- **SAFETY FILTER**: Validate food and donor
- **URGENCY + DISTANCE + CAPACITY**: Prioritize by time, distance and NGO capacity
- **FEASIBLE MATCH**: Find the best available volunteer or NGO
- **PICKUP**: Confirm pickup and start tracking

THE FOODBRIDGE ADVANTAGE

- **Faster feasible matches** that save more food
- **Better coordination** across donors, NGOs and volunteers
- **Fewer missed pickups** through deadline-aware alerts
- **Transparent records** of food and impact
- **Repeatable process** that scales across communities

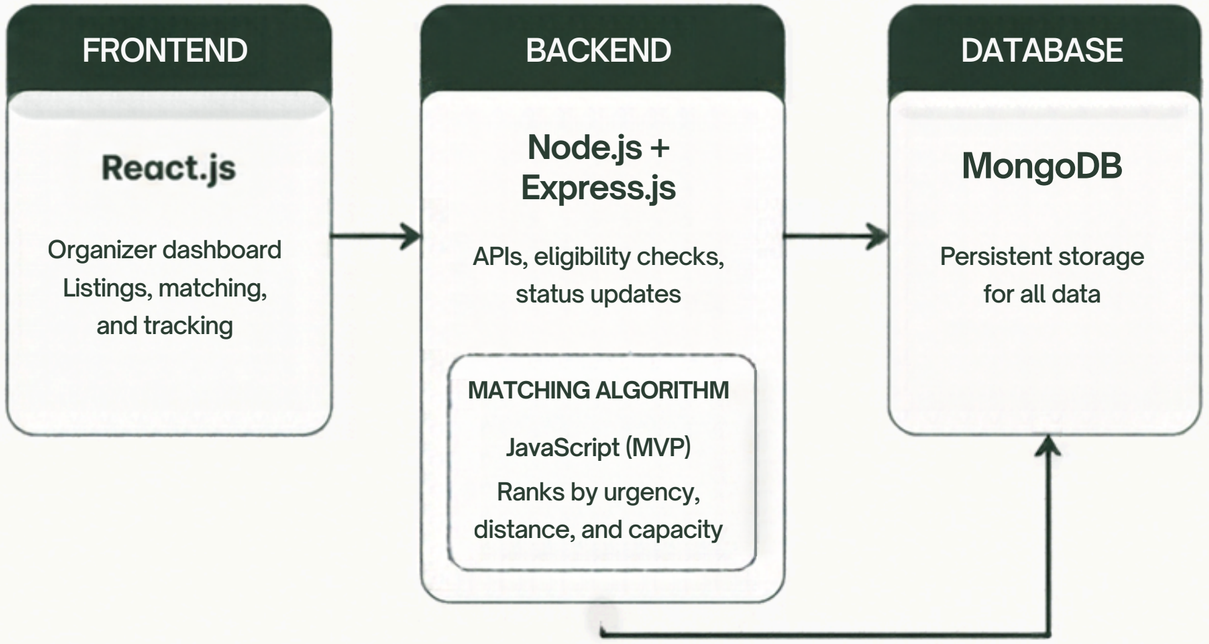
SAFETY-FIRST MATCHING IS OUR CORE INNOVATION

TECHNOLOGY STACK

The technology powering FoodBridge

React.js	Organizer dashboard for creating listings, matching recipients, and tracking pickups.
node + express	APIs, eligibility checks, matching logic, and real-time status updates.
MongoDB	Stores donor, event, food, capacity, deadlines, and pickup records.
JS	Filters and ranks recipients by urgency, distance, and capacity.
Google Maps API / Mock coordinates	Distance calculations and mapping support.
Firebase Auth + Cloud Messaging	Authentication and notifications (simulated for demo).

ARCHITECTURE OVERVIEW



Implemented (MVP)
Core platform built and tested

Planned / Simulated
Integrations for demo and future release

WORKFLOW + PROTOTYPE

From Event Surplus to Community Meals

01
List Food
Organizer lists surplus meals with quantity, time, and location.

02
Check Eligibility
System applies food safety and timing rules.

03
Match Recipients
Algorithm ranks NGOs by urgency, distance, and capacity.

04
Send Alert
Suitable NGOs receive alert and respond to pickup.

05
Track Pickup
Pickup moves through Accepted → Collected → Delivered.

06
Record Impact
Dashboard logs meals redistributed and food quantities saved.

ORGANIZER DASHBOARD

Sample Wedding Reception

Surplus Portions
40 meals

Prepared At
1:00 PM

Collection Deadline
3:00 PM

Location
Delhi, India

Post Surplus

SMART MATCHING RESULTS

Feasible Match for 40 meals

NGO A

Distance
2 km

Capacity
50 meals

Match Status
Feasible
Can accept 40 meals

Accept Pickup

NGO B

Distance
1 km

Capacity
10 meals

Match Status
Insufficient
Can accept 10 meals

Matching based on urgency, distance, capacity, and safety window.

IMPACT DASHBOARD

Confirmed Meals Redistributed
40 meals

Food Rescued
18 kg

Donations Delivered
1 Pickup

Estimated Food Waste Avoided
25 kg CO_2e

Real Impact. Measurable Change.

FoodBridge turns surplus into support by getting the right food and ad the right people before its safe collection window closes.